



Centar izvrsnosti
Varaždin Croatia
Team

Soccer 1:1
infrared
league



About our city

We live in Varaždin. Varaždin is a city in northern Croatia. It is right next to the river Drava and it's a historical, cultural, educational, economic, sports and tourist center of Varaždin County.

Data / Results / Discussion

So far our robot is working as expected, while having a few hiccups every now and then because of the sensors. Our robot doesn't move the fastest but we are trying to improve the speed. The so called "hiccups" mostly used to happen when the robot would hit a wall and the sensors couldn't see the ball, we fixed this problem by making things like the handle tinner to wide and moving the sensors outwards. We also made the robot move smoother by using more complex and smooth wheels.

Method / Robot production / Design

After we qualifiled for the turnament we desided to redesign our robot from the ground up. We desided to use ultrasonic sensors and much better motors and wheels then we made a chassis that has removable walls for easier production.

Abstract

We are a team of four primary school students from Varaždin, this is first year as a team, we have been working on our robot since Februrary after the national competition, our robot has broken down multiple times. The programming is made in C++. Our robot is equipped with a dribbler.

Software we use

We use Autodesk fusion 360 for our CAD models, Arduino IDE for our robots program and Affinity for our poster.

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ESP32

LED matrix

3 motors

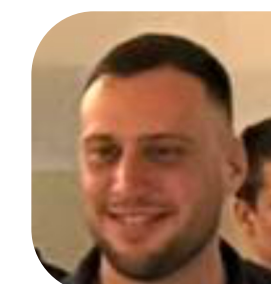
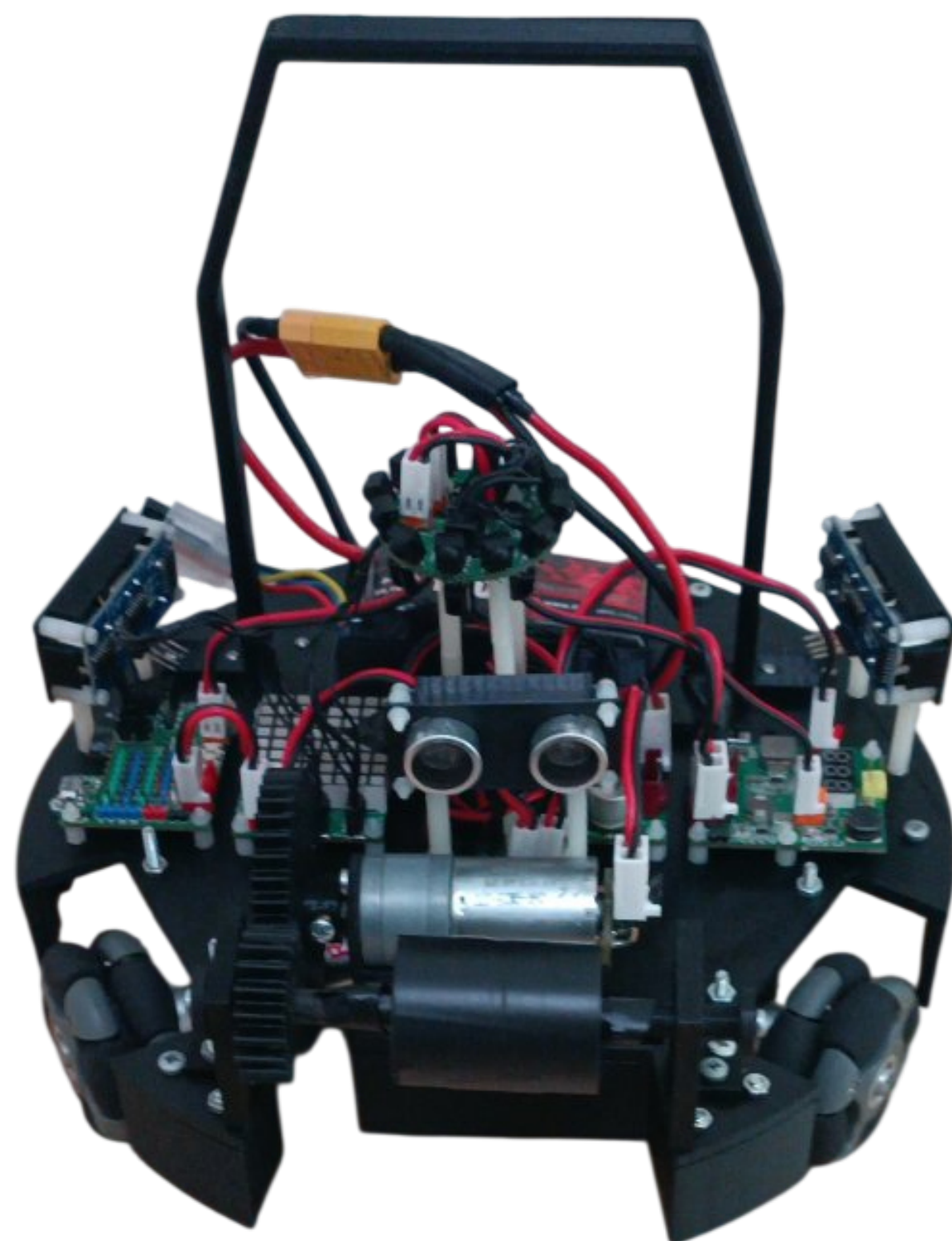
Power supply

Battery

Motor controller

Transformer

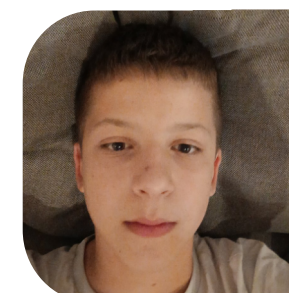
Dribbler



Ivan Kolarić:
Team mentor
(not member)



Leon Rođak:
Team captain,
robot 3D designer



Pavao Margotič:
Team member,
software



Tomislav Runjić:
Team member,
documentation



Julian Erik Cupek:
Team member,
robot assembler